

Imagine a scenario a few years into the future

You wake up in the morning to the aroma of breakfast wafting through the house. You groggily walk down the stairs wondering, “Who’s up so early and slaving away in the kitchen to prepare my favourite pancakes?” Nope it’s the ‘Replicator 9000’, the bio-printing device you got yourself a few days ago. It beeps to notify you that the freshly made (or rather, freshly printed) pancakes are steadily rolling out of it. And this is just the beginning of your day. Besides printing food, 3D Printing a few years into the future will also enable the production of living tissue (so no more waiting for donor matches for organs), ready made electronics, customised clothing from shoes to belt buckles and perhaps even an entire aircraft.

The day is not too far away now. 3D Printing, in its early stages, is already achieving many great feats – manufacturing miracles if you may. It is one of the most important disruptive technologies to take the world by storm. We’re a lucky generation to be witnessing this paradigm shift first hand. Some are hailing it the second coming of the industrial revolution where the power to make and create will no longer be concentrated in the hands of a few but rather in the hands of the masses.

In the chapters that follow we delve into this upcoming technology in depth. We begin by explaining what 3D Printing is, its history and some of its possible implications. For the more technically inclined, we then look at how 3D Printers work. Those of you wanting to put this technical knowledge to practical use are in luck. If you have the resourcefulness you can even make your own 3D printer. Granted it’s not as easy as it looks, but we’ve tried to simplify the process for you. Add to the fact that the power of Open Source is involved. With designs for printers as well as the objects to be printed themselves easily available, the power is indeed in the hands of the people.

The most important implication of 3D printing in the future is the empowerment of ordinary people create what they want, when they want and however they want. Let’s get started... 